

座位号

专业

学院

学号

姓名

诚信应考,考试作弊将带来严重后果!

华南理工大学期末考试

2023 Calculus Mid-Term Exam

- 注意事项: 1. 考前请将密封线内填写清楚;
2. 所有答案请直接答在试卷上;
3. 考试形式: 闭卷;
4. 本试卷共 9 大题, 满分 100 分, 考试时间 120 分钟。

题 号	1, 2	3, 4	5, 6	7, 8	9	总分
得 分						
评卷人						

1. Answer the questions (30):

(1) Does the series $\sum_{n=1}^{\infty} \left(\frac{\sin na}{n^2} - \frac{1}{\sqrt{n}} \right)$ converge or diverge?

Diverge. The series

$$\sum_{n=1}^{\infty} \frac{\sin na}{n^2} \leq \sum_{n=1}^{\infty} \left| \frac{\sin na}{n^2} \right| \leq \sum_{n=1}^{\infty} \left| \frac{1}{n^2} \right|$$

converge. But $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ diverge. So the series $\sum_{n=1}^{\infty} \left(\frac{\sin na}{n^2} - \frac{1}{\sqrt{n}} \right)$ diverge

(2) Find the volume of the parallelepiped with edges $3\vec{i} - 4\vec{j} + 2\vec{k}$, $-\vec{i} + 2\vec{j} + \vec{k}$ and $3\vec{i} - 2\vec{j} + 5\vec{k}$

The volume $V = |(a_1 \times a_2) \cdot a_3|$, where $a_1 = 3\vec{i} - 4\vec{j} + 2\vec{k}$, $a_2 = -\vec{i} + 2\vec{j} + \vec{k}$,

$$a_3 = 3\vec{i} - 2\vec{j} + 5\vec{k}. \quad V = \left| (3\vec{i} - 4\vec{j} + 2\vec{k}) \cdot (3\vec{i} - 2\vec{j} + 5\vec{k}) \right| = |-24 + 10 + 10| = 4$$

(密封线内不答题)

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(3) Suppose $x^3 e^{y+z} - y \cos(2xz) = 0$, find $\partial z / \partial x$

Suppose z is a function of x , $f = x^3 e^{y+z} - y \cos(2xz)$.

$$\frac{\partial f}{\partial x} = 3x^3 e^{y+z} + x^3 e^{y+z} \frac{\partial z}{\partial x} + y \sin 2xz \cdot 2 \left(z + x \frac{\partial z}{\partial x} \right) = 0$$

$$\frac{\partial z}{\partial x} (x^3 e^{y+z} + 2xy \sin 2xz) = -3x^3 e^{y+z} - 2yz \sin 2xz$$

$$\frac{\partial z}{\partial x} = -\frac{3x^3 e^{y+z} + 2yz \sin 2xz}{x^3 e^{y+z} + 2xy \sin 2xz}$$

(4) Let $z = f\left(2x, \frac{y^2}{x}\right)$ has the second-order continuous partial derivatives, find

$$\frac{\partial^2 z}{\partial x \partial y}.$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{-2y^2}{x^2} f_x \left(2x, \frac{y^2}{x} \right) \right)$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{-4y^3}{x^3} f_{xy} \left(2x, \frac{y^2}{x} \right) + \frac{-4y}{x^2} f_x \left(2x, \frac{y^2}{x} \right)$$

(5) Find the equation of the plane through $(-1, -2, 3)$ and perpendicular to both the plane $x - 3y + 2z = 7$ and $2x - 2y - z = -3$

Set the plane is $ax + by + cz = d$, and it satisfies the follow:

$$\begin{cases} -a - 2b + 3c = d \\ a - 3b + 2c = 0 \\ 2a - 2b - c = 0 \end{cases}$$

Then we have

$$\begin{cases} a = \frac{7}{5}b \\ c = \frac{4}{5}b \\ d = -b \end{cases}$$

So the plane is $7x + 5y + 4z = -5$

(6) Find ∇f and $D_u f$, where $f = x^3 + \tan yz$, $u = (2, 0, -4)$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \left(3x^2, \frac{z}{\cos^2 yz}, \frac{y}{\cos^2 yz} \right)$$

Unitize $u = (2, 0, -4)$, $\tilde{u} = \left(\frac{1}{\sqrt{5}}, 0, -\frac{2}{\sqrt{5}} \right)$

$$D_u f = \left(3x^2, \frac{z}{\cos^2 yz}, \frac{y}{\cos^2 yz} \right) \cdot \left(\frac{1}{\sqrt{5}}, 0, -\frac{2}{\sqrt{5}} \right) = \frac{1}{\sqrt{5}} \left(3x^2 - \frac{2y}{\cos^2 yz} \right)$$

2. Evaluate the problems (30):

(1) Rewrite the iterated integral with the indicated order of integration

$$\int_{-1}^0 \int_{-\sqrt{y+1}}^{\sqrt{y+1}} f(x, y) dx dy, \quad dy dx$$

$$\int_{-1}^0 \int_{-\sqrt{y+1}}^{\sqrt{y+1}} f(x, y) dx dy = \int_{-1}^1 \int_{x^2-1}^0 f(x, y) dy dx$$

(2) Switch to rectangular coordinate and evaluate $\int_{3\pi/4}^{4\pi/3} d\theta \int_0^{-5\sec\theta} r^3 \sin\theta dr$

Set:

$$x = r \cos \theta, \quad y = r \sin \theta,$$

$$\int_{\frac{3\pi}{4}}^{\frac{4\pi}{3}} d\theta \int_0^{-5\sec\theta} r^3 \sin\theta dr = \int_{-5}^0 dx \int_{\sqrt{3}x}^{-x} y \sqrt{x^2 + y^2} dy$$

Evaluate:

$$\begin{aligned} \int_{-5}^0 dx \int_{\sqrt{3}x}^{-x} y \sqrt{x^2 + y^2} dy &= \int_{-5}^0 \frac{1}{3} (x^2 + y^2)^{\frac{3}{2}} \bigg|_{\sqrt{3}x}^{-x} dx \\ &= \int_{-5}^0 \frac{2\sqrt{2}-8}{3} x^3 dx \\ &= \frac{(8-2\sqrt{2})625}{12} \end{aligned}$$

(3) Switch the order and then evaluate the integrals:

$$\int_{\frac{1}{4}}^{\frac{1}{2}} dy \int_{\frac{1}{2}}^{\sqrt{y}} e^{\frac{y}{x}} dx + \int_{\frac{1}{2}}^1 dy \int_y^{\sqrt{y}} e^{\frac{y}{x}} dx$$

Switch:

$$\begin{aligned} \int_{\frac{1}{4}}^{\frac{1}{2}} dy \int_{1/2}^{\sqrt{y}} e^{\frac{y}{x}} dx + \int_{\frac{1}{2}}^1 dy \int_y^{\sqrt{y}} e^{\frac{y}{x}} dx &= \int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} dx \int_{x^2}^{\frac{1}{2}} e^{\frac{y}{x}} dy + \int_{\frac{1}{\sqrt{2}}}^1 dx \int_{x^2}^x e^{\frac{y}{x}} dy + \int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} dx \int_{\frac{1}{2}}^x e^{\frac{y}{x}} dy \\ &= \int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} dx \int_{x^2}^x e^{\frac{y}{x}} dy + \int_{\frac{1}{\sqrt{2}}}^1 dx \int_{x^2}^x e^{\frac{y}{x}} dy \\ &= \int_{\frac{1}{2}}^1 dx \int_{x^2}^x e^{\frac{y}{x}} dy \end{aligned}$$

Evaluate

$$\begin{aligned} \int_{\frac{1}{2}}^1 dx \int_{x^2}^x e^{\frac{y}{x}} dy &= \int_{\frac{1}{2}}^1 \frac{x}{2} e^{\frac{y}{x}} \bigg|_{x^2}^x dx \\ &= \int_{\frac{1}{2}}^1 \left(\frac{x}{2} e - \frac{x}{2} e^x \right) dx \\ &= \frac{3e}{16} - \frac{1}{4} e^{\frac{1}{2}} \end{aligned}$$

(4) Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{1}{\sqrt{4-x^2-y^2}} dy dx$

Set:

$$x = r \cos \theta, \quad y = r \sin \theta,$$

then

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{1}{\sqrt{4-x^2-y^2}} dy dx = \int_0^{\frac{\pi}{2}} d\theta \int_0^1 \frac{1}{\sqrt{4-r^2}} r dr = \frac{\pi}{2} (2 - \sqrt{3})$$

(5) Find $\iint_D \frac{y}{x+y} e^{(x+y)^2} dx dy$, D is bounded by $x+y=1$, $x=0$ and $y=0$.

Set $u = x+y$, $v = x-y$, then

$$x = \frac{u+v}{2}, \quad y = \frac{u-v}{2}$$

$$dx dy = \frac{1}{2} du dv$$

$$\begin{aligned} \iint_D \frac{y}{x+y} e^{(x+y)^2} dx dy &= \int_0^1 du \int_{-u}^u \frac{u-v}{4u} e^{u^2} dv \\ &= \frac{1}{4} \int_0^1 2ue^2 du \\ &= \frac{e-1}{4} \end{aligned}$$

(6) Compute the triple integral $\iiint_{\Omega} z dv$, where $\Omega: \begin{cases} x^2 + y^2 + z^2 \leq 2 \\ z \geq x^2 + y^2 \end{cases}$

Set

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z,$$

then

$$dv = r dr d\theta dz$$

$$\begin{aligned}
\iiint_{\Omega} z dv &= \int_0^{2\pi} d\theta \int_0^1 dr \int_{r^2}^{\sqrt{2-r^2}} z r dz \\
&= 2\pi \int_0^1 \frac{r}{2} (2 - r^2 - r^4) dr \\
&= \frac{7}{12} \pi
\end{aligned}$$

3 (5) Find the minimum and maximum for the function $z^2 = x^2 + y^2$ with the constraint $(x - \sqrt{2})^2 + (y - \sqrt{2})^2 \leq 9$.

Set

$$L = x^2 + y^2 + \lambda \left((x - \sqrt{2})^2 + (y - \sqrt{2})^2 - 9 \right)$$

then

$$L_x = 2x + 2\lambda(x - \sqrt{2}) = 0$$

$$L_y = 2y + 2\lambda(y - \sqrt{2}) = 0$$

$$x = y = \frac{\sqrt{2}\lambda}{1 + \lambda}$$

$$\left(\frac{\sqrt{2}\lambda}{1 + \lambda} - \sqrt{2} \right)^2 + \left(\frac{\sqrt{2}\lambda}{1 + \lambda} - \sqrt{2} \right)^2 \leq 9$$

$$\lambda \geq -\frac{1}{3} \text{ or } \lambda \leq -\frac{5}{3}$$

$$f = x^2 + y^2 = \frac{4\lambda^2}{(1 + \lambda)^2} \geq 0$$

$$\frac{4\lambda^2}{(1 + \lambda)^2} = \frac{4}{\left(1 + \frac{1}{\lambda}\right)^2} \leq 9$$

The minimum is 0, the maximum is 9.

4 (5) Find convergence set and sum for the power series $\sum_{n=1}^{\infty} n(n+1)(x-1)^n$

$$\lim_{n \rightarrow \infty} \sqrt[n]{n(n+1)} = \lim_{n \rightarrow \infty} \sqrt[n]{n^2} = 1$$

$$R = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{n(n+1)}} = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{n^2}} = \frac{1}{1} = 1$$

Hence the convergence set is

$$\{x \in \mathbb{R} \mid 0 < x < 2\}$$

Differentiating the series for $\frac{1}{x-2}$ twice yields

$$\frac{2}{(2-x)^3} = 2 + 6(x-1) + 12(x-1)^2 + 20(x-1)^3 + \dots$$

Multiplying this series by $x-1$ gives

$$\frac{2(x-1)}{(2-x)^3} = 2(x-1) + 3 \cdot 2(x-1)^2 + 4 \cdot 3(x-1)^3 + 5 \cdot 4(x-1)^4 + \dots$$

$$\text{hence } \sum_{n=1}^{\infty} n(n+1)(x-1)^n = \frac{2(x-1)}{(2-x)^3} \text{ for } 0 < x < 2.$$

$$5 \text{ (5) If } f(x, y) = \begin{cases} \frac{x^3 y}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}, \text{ find } f_{xy}(0, 0).$$

$$f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = 0, \quad y = 0$$

$$f_x(0, y) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, y) - f(0, y)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x^2 y}{\Delta x^2 + y^2} = 0, \quad y \neq 0$$

$$f_{xy}(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{f_x(0, \Delta y) - f_x(0, 0)}{\Delta y} = 0$$

$$6 \text{ (5) Calculate } \int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^2 (x^2 + y^2)^{1/2} dz dy dx$$

Set

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z,$$

then

$$dzdydx = r dr d\theta dz$$

$$\begin{aligned}\int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^2 (x^2 + y^2)^{\frac{1}{2}} dz dy dx &= \int_0^2 dz \int_0^{\frac{\pi}{2}} d\theta \int_0^3 r^2 dr \\ &= 2 \cdot \frac{\pi}{2} \cdot 9 \\ &= 9\pi\end{aligned}$$

7 (5) Find the parametric equation of the line that is tangent to the curve of intersection of the surface $x = z^2$ and $y = z^3$ at $(1, 1, 1)$

Set

$$F = x - z^2, \quad G = y - z^3,$$

so the normal vector of them at $(1, 1, 1)$ is

$$n_1 = (1, 0, -2), \quad n_2 = (0, 1, -3)$$

Then the line parallel to $n_1 \times n_2$

$$n_1 \times n_2 = (2, 3, 1)$$

Hence the parametric equation of the line is

$$\begin{cases} x = 2t + 1 \\ y = 3t + 1 \\ z = t + 1 \end{cases}$$

8 (5) Find the volume of the solid bounded by the surface $z = xy$ and planes

$$z = x + y, \quad x + y = 1, \quad x = 0, \quad y = 0.$$

$$\begin{aligned}
V &= \int_0^1 dx \int_0^{1-x} dy \int_{xy}^{x+y} dz \\
&= \int_0^1 dx \int_0^{1-x} (x+y-xy) dy \\
&= \frac{1}{2} \int_0^1 (-x^3 + x^2 - x + 1) dx \\
&= \frac{7}{24}
\end{aligned}$$

9. (5) Consider the Cobb-Douglas production model for a manufacturing process depending on three input x, y, z with unit costs a, b, c , respectively. Given by

$$P(x, y, z) = k x^\alpha y^\beta z^\gamma \quad \alpha > 0, \beta > 0, \gamma > 0, \quad \alpha + \beta + \gamma = 1$$

subject to the cost constrain $ax + by + cz = 1$. Determine the x, y, z maximize the production

Without loss of generality $k > 0, x > 0, y > 0, z > 0$, then

$$\log P(x, y, z) = \log k + \alpha \log x + \beta \log y + \gamma \log z$$

So the x, y, z maximize $\log P(x, y, z)$ also maximize $P(x, y, z)$

Set

$$L = \log k + \alpha \log x + \beta \log y + \gamma \log z - \lambda (ax + by + cz - 1)$$

then

$$L_x = \frac{\alpha}{x} - \lambda a = 0$$

$$L_y = \frac{\beta}{y} - \lambda b = 0$$

$$L_z = \frac{\gamma}{z} - \lambda c = 0$$

$$x = \sqrt{\frac{\alpha}{\lambda a}}, y = \sqrt{\frac{\beta}{\lambda b}}, z = \sqrt{\frac{\gamma}{\lambda c}}$$

$$\begin{aligned}
\lambda &= \lambda ax + \lambda by + \lambda cz \\
&= \frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} \\
&= \sqrt{\lambda \alpha a} + \sqrt{\lambda \beta b} + \sqrt{\lambda \gamma c}
\end{aligned}$$

The x, y, z maximize the production is the follow:

$$x = \frac{\sqrt{\alpha}}{\sqrt{a}(\sqrt{\alpha a} + \sqrt{\beta b} + \sqrt{\gamma c})}, y = \frac{\sqrt{\beta}}{\sqrt{b}(\sqrt{\alpha a} + \sqrt{\beta b} + \sqrt{\gamma c})}, z = \frac{\sqrt{\gamma}}{\sqrt{c}(\sqrt{\alpha a} + \sqrt{\beta b} + \sqrt{\gamma c})}$$

10.(5) Evaluate $\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{-\sqrt{9-x^2-z^2}}^{\sqrt{9-x^2-z^2}} (x^2 + y^2 + z^2)^{3/2} dy dz dx$

Set

$$x = r \sin \theta \cos \varphi, \quad y = r \cos \theta, \quad z = r \sin \theta \sin \varphi$$

Then

$$dy dz dx = r^2 \sin \theta dr d\theta d\varphi$$

$$\begin{aligned}
\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{-\sqrt{9-x^2-z^2}}^{\sqrt{9-x^2-z^2}} (x^2 + y^2 + z^2)^{3/2} dy dz dx &= \int_0^{\pi} d\theta \int_0^{2\pi} d\varphi \int_0^3 r^5 \sin \theta dr \\
&= 243\pi \int_0^{\pi} \sin \theta d\theta \\
&= 486\pi
\end{aligned}$$